

GATE DATASHEET

8-Bit Counter Design

Field	Value
Course	ED222 — Digital IC Design Tape out Lab
Main library	8BitCounter_PHY_ED222
Gates library	Gates_ED222_PY
Top module	8Bit_Counter_Integrated
Tool	Cadence Virtuoso
Technology	180nm CMOS
Supply voltage	1.8 V

Group – 4

Members

Prince Kirti Patel: 202404031

Yash Popat: 202404047

Hemin Sidhhapura: 202404055

Table of Contents

1. Test Conditions & Library Overview	4
Simulation Environment	4
Library Structure	4
2. NAND Gate	5
Overview	5
Pin Configuration	5
Truth Table	5
Schematic/Layout	6
Timing Analysis	6
Propagation Delay	6
Design Notes	6
3. NOT Gate (Inverter) — NOT_with_NAND	7
Overview	7
Pin Configuration	7
Truth Table	7
Layout/Schematic	8
Timing Analysis	8
Propagation Delay	8
Design Notes	8
4. AND Gate — AND_with_NAND	9
Overview	9
Pin Configuration	9
Truth Table	9
Layout/Schematic	10
Propagation Delay	10
Design Notes	10
5. OR Gate — OR_with_NAND	11
Overview	11
Pin Configuration	11
Truth Table	11
Layout/Schematic	12
Timing Analysis	12
Propagation Delay	12
Design Notes	12
6. EXOR Gate — EXOR_with_NAND	13

Overview	13
Pin Configuration	13
Truth Table	13
Schematic	14
Propagation Delay	15
Design Notes	15
7. D Flip-Flop — D_FF_with_NAND	16
Overview	16
Pin Configuration	16
Characteristic Table	16
Schematic	17
Propagation Delay	18
Design Notes	18

1. Test Conditions & Library Overview

Simulation Environment

Parameter	Value
Simulator	Cadence
Analysis type	Transient
Supply voltage	1.8 V
Input voltage LOW	0 V
Input voltage HIGH	1.8 V
Measurement (tpd)	50% VDD crossing (0.9 V)
Measurement (tr/tf)	10% to 90% VDD (0.18 V to 1.62 V)

Library Structure

All gates in **Gates_ED222_PY** are implemented entirely from the NAND gate at transistor level. No PDK standard cells are used.

Cell name	Function	Used in module
NAND	2-input NAND	D_FF_with_NAND, AND_with_NAND, EXOR_with_NAND, OR_with_NAND, NOT_with_NAND
NOT_with_NAND	Inverter	D_FF_with_NAND, OR_with_NAND, PISO_SR
AND_with_NAND	2-input AND	8Bit_Synchronous_Counter, PISO_SR, D_FF_with_NAND
OR_with_NAND	2-input OR	PISO_SR
EXOR_with_NAND	2-input EXOR	8Bit_Synchronous_Counter
D_FF_with_NAND	D Flip-Flop (M-S)	8Bit_Synchronous_Counter, PISO_SR

2. NAND Gate

Overview

Parameter	Value
Full cell name	NAND
Boolean expression	$Y = (A \cdot B)'$
Implementation	Direct CMOS — 2 PMOS (parallel) + 2 NMOS (series)
Library	Gates_ED222_PY
Technology	80nm CMOS
Supply voltage (VDD)	1.8 V

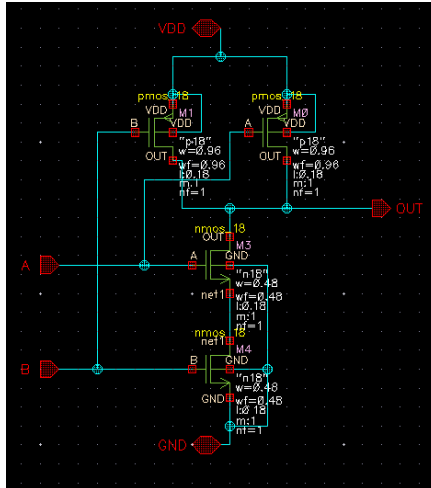
Pin Configuration

Pin	Direction	Type	Description
A	Input	Logic	First data input
B	Input	Logic	Second data input
Y	Output	Logic	NAND output — HIGH unless both inputs HIGH
VDD	Power	Power	Supply voltage — 1.8 V
GND	Power	Power	Ground reference — 0 V

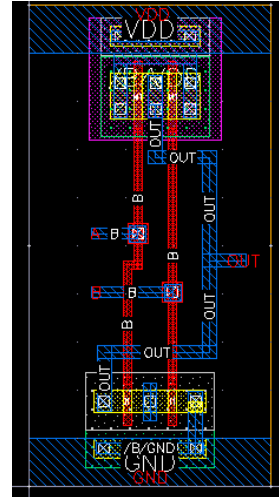
Truth Table

A	B	$Y = (A \cdot B)'$
0	0	1
0	1	1
1	0	1
1	1	0

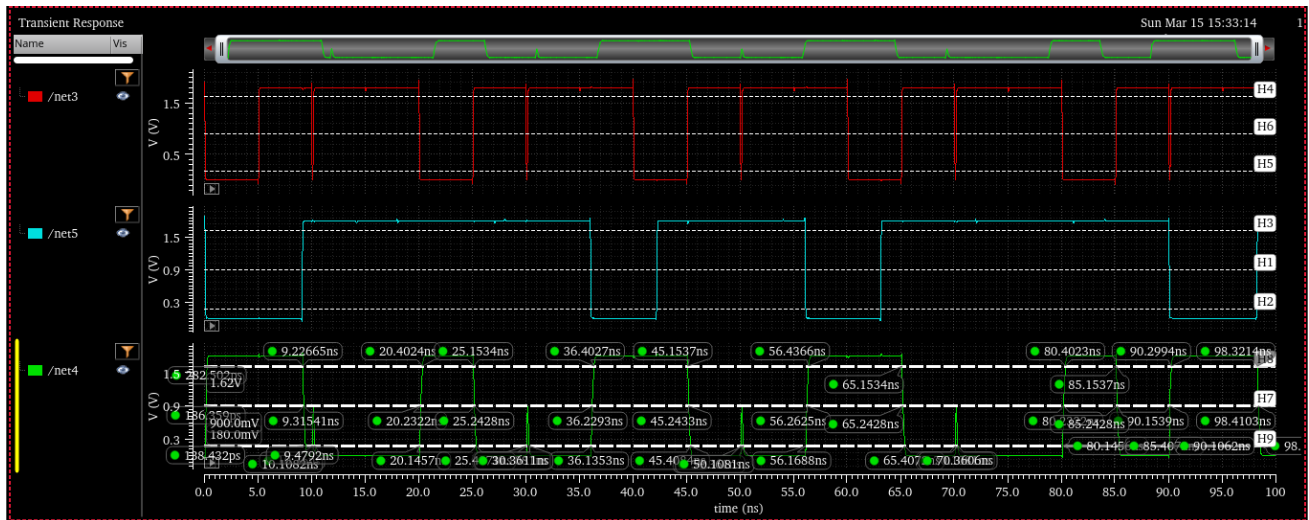
Schematic



Layout



Timing Analysis



Propagation Delay

Parameter	Symbol	Condition (one input constant)	Value	Unit
Propagation delay HL	t_{pHL}	Y: 1→0	0.1075	ns
Propagation delay LH	t_{pLH}	Y: 0→1	0.1333	ns
Average prop. delay	t_p	$(t_{pHL} + t_{pLH}) / 2$	0.1204	ns

Design Notes

- Base cell of the entire library — all other gates are derived from this cell.
- Series NMOS transistors in the PDN are sized at $W = 0.48 \mu\text{m}$.
- Parallel PMOS transistors in the PUN are sized at $W = 0.96 \mu\text{m}$.
- t_{pHL} is typically shorter than t_{pLH} due to higher NMOS drive current (electron vs. hole mobility).

3. NOT Gate (Inverter) — NOT_with_NAND

Overview

Parameter	Value
Full cell name	NOT_with_NAND
Boolean expression	$Y = A'$
Implementation	NAND gate with both inputs tied together: $Y = \text{NAND}(A, A)$
Library	Gates_ED222_PY
Technology	180nm CMOS
Supply voltage (VDD)	1.8 V

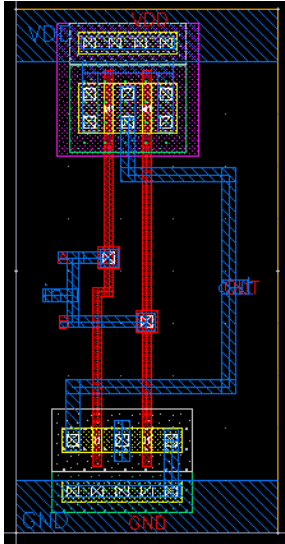
Pin Configuration

Pin	Direction	Type	Description
A	Input	Logic	Data input
Y	Output	Logic	Inverted output
VDD	Power	Power	Supply voltage — 1.8 V
GND	Power	Power	Ground reference — 0 V

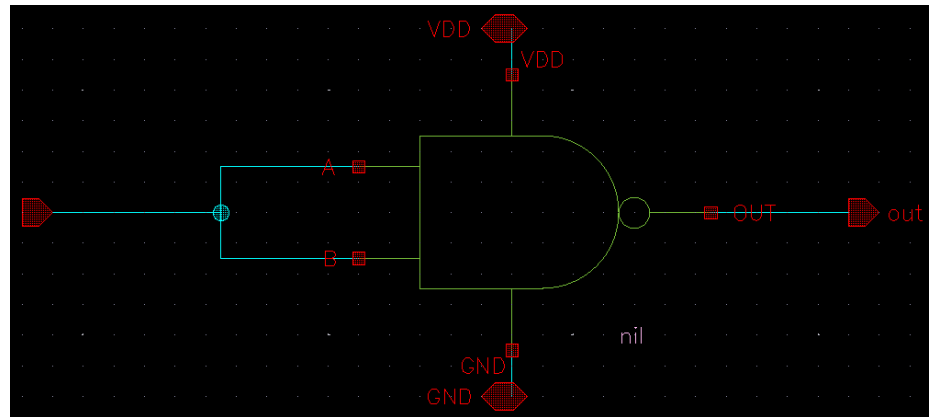
Truth Table

A	$Y = A'$
0	1
1	0

Layout



Schematic



Timing Analysis



Propagation Delay

Parameter	Symbol	Condition (one input constant)	Value	Unit
Propagation delay HL	t_{pHL}	Y: 1→0	0.0459	ns
Propagation delay LH	t_{pLH}	Y: 0→1	0.0617	ns
Average prop. delay	t_p	$(t_{pHL} + t_{pLH}) / 2$	0.0538	ns

Design Notes

- Implemented using a single NAND gate with both inputs shorted to signal A.
- No additional transistors required — reuses the NAND cell with a wiring modification.
- Used internally in D_FF_with_NAND for clock inversion and feedback paths.

4. AND Gate — AND_with_NAND

Overview

Parameter	Value
Full cell name	AND_with_NAND
Boolean expression	$Y = A \cdot B$
Implementation	NAND gate followed by NOT (inverter): $Y = \text{NOT}(\text{NAND}(A, B))$
Library	Gates_ED222_PY
Technology	180nm CMOS
Supply voltage (VDD)	1.8 V

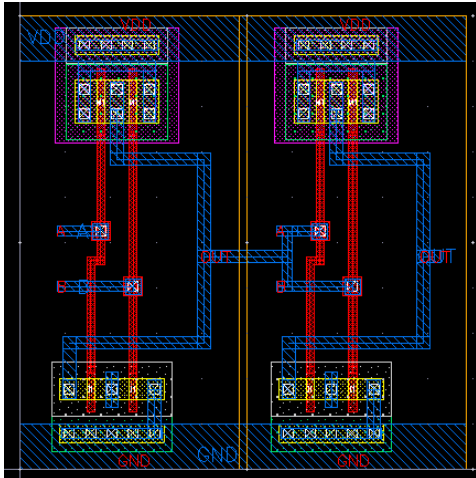
Pin Configuration

Pin	Direction	Type	Description
A	Input	Logic	First data input
B	Input	Logic	Second data input
Y	Output	Logic	AND output — HIGH only when both inputs HIGH
VDD	Power	Power	Supply voltage — 1.8 V
GND	Power	Power	Ground reference — 0 V

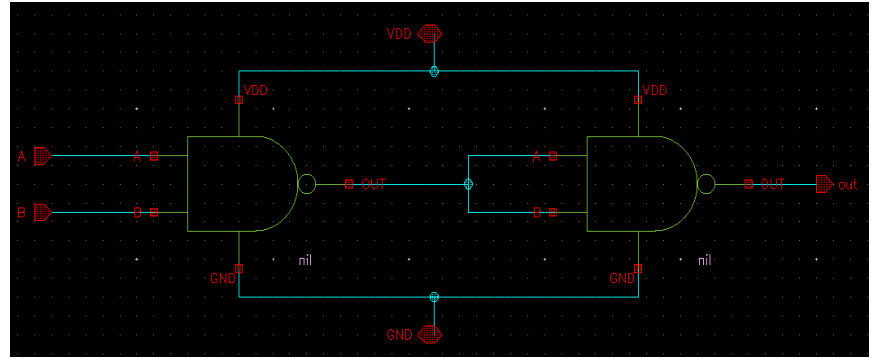
Truth Table

A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

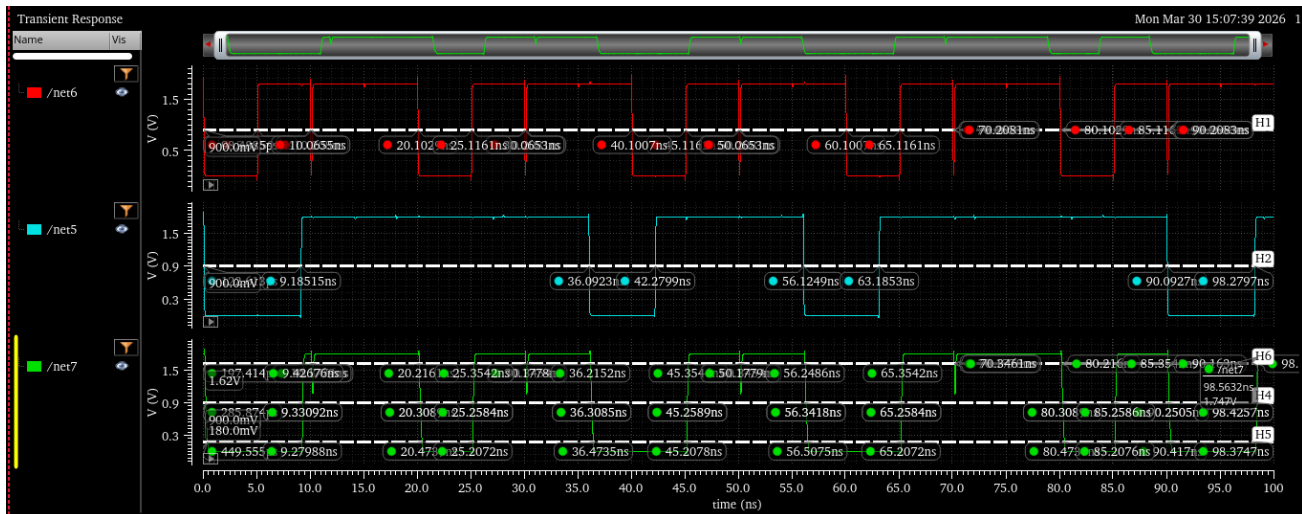
Layout



Schematic



Timing Analysis



Propagation Delay

Parameter	Symbol	Condition (one input constant)	Value	Unit
Propagation delay HL	t _{pHL}	Y: 0→1	0.2111	ns
Propagation delay LH	t _{pLH}	Y: 1→0	0.1441	ns
Average prop. delay	t _p	(t _{pHL} + t _{pLH}) / 2	0.1776	ns

Design Notes

- Implemented as two cascaded cells: NAND → NOT_with_NAND.
- Propagation delay is approximately t_p(NAND) + t_p(NOT).
- Used in: 8Bit_Synchronous_Counter (carry chain enable), PISO_SR (load control), D_FF_with_NAND (internal).

5. OR Gate — OR_with_NAND

Overview

Parameter	Value
Full cell name	OR_with_NAND
Boolean expression	$Y = A + B$
Implementation	De Morgan equivalent: $Y = \text{NAND}(A', B') = \text{NAND}(\text{NOT}(A), \text{NOT}(B))$
Library	Gates_ED222_PY
Technology	180nm CMOS
Supply voltage (VDD)	1.8 V

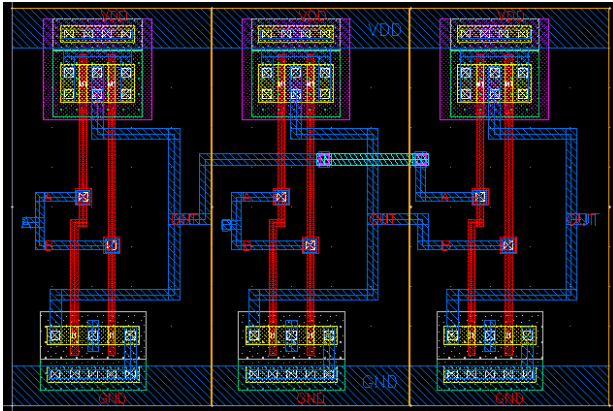
Pin Configuration

Pin	Direction	Type	Description
A	Input	Logic	First data input
B	Input	Logic	Second data input
Y	Output	Logic	OR output — LOW only when both inputs LOW
VDD	Power	Power	Supply voltage — 1.8 V
GND	Power	Power	Ground reference — 0 V

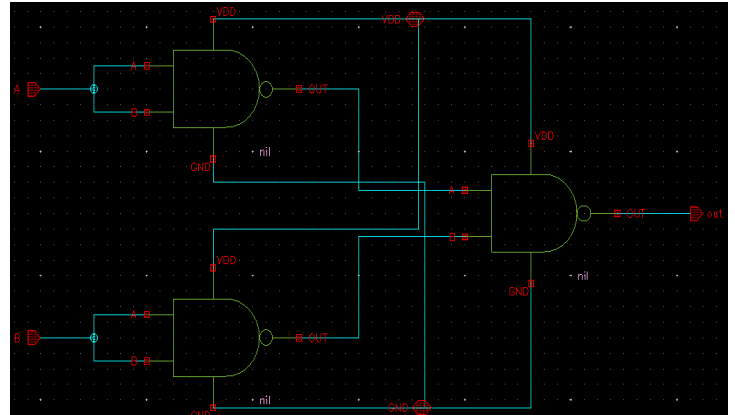
Truth Table

A	B	Y = A+B
0	0	0
0	1	1
1	0	1
1	1	1

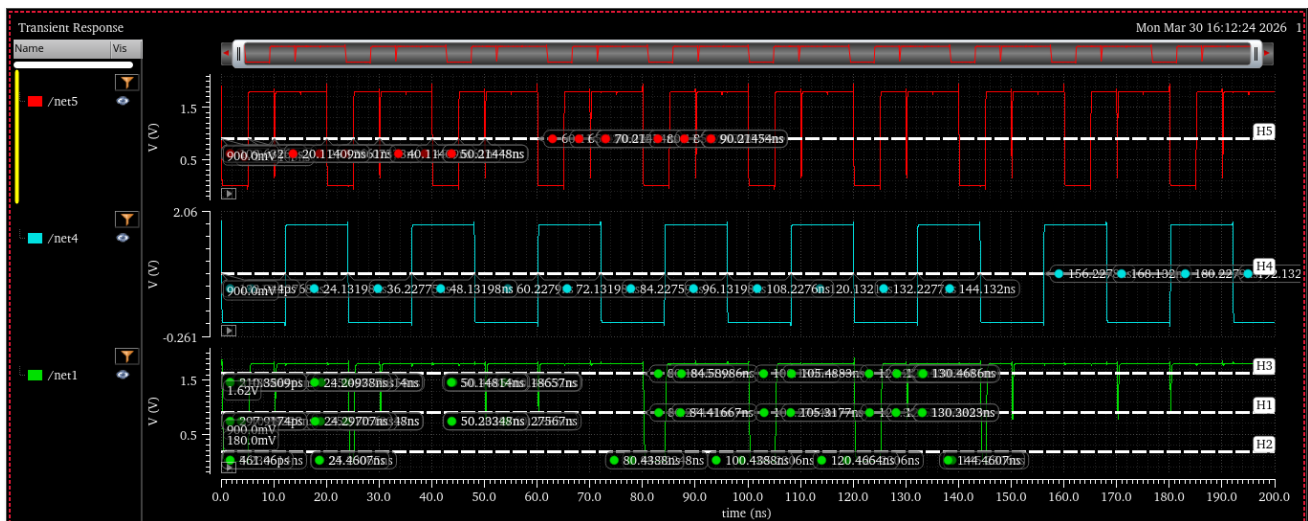
Layout



Schematic



Timing Analysis



Propagation Delay

Parameter	Symbol	Condition (one input constant)	Value	Unit
Propagation delay HL	t _{pHL}	Y: 1→0	0.1626	ns
Propagation delay LH	t _{pLH}	Y: 0→1	0.1904	ns
Average prop. delay	t _p	(t _{pHL} + t _{pLH}) / 2	0.1765	ns

Design Notes

- Implemented using De Morgan's theorem: $A+B = (A' \cdot B')$ = NAND(NOT(A), NOT(B)).
- Uses three NAND-family cells: 2× NOT_with_NAND + 1× NAND.
- Used in PISO_SR for MUX select logic.
- t_{pLH} is higher than other gates due to three cascaded logic levels.

6. EXOR Gate — EXOR_with_NAND

Overview

Parameter	Value
Full cell name	EXOR_with_NAND
Boolean expression	$Y = A \oplus B = A'B + AB'$
Implementation	4-NAND topology — all derived from the NAND base cell
Library	Gates_ED222_PY
Technology	180nm CMOS
Supply voltage (VDD)	1.8 V

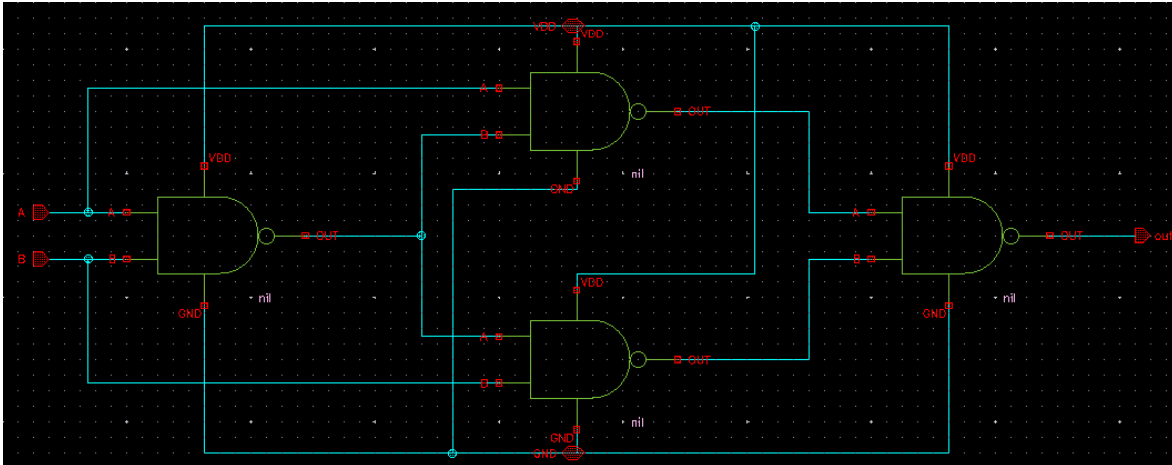
Pin Configuration

Pin	Direction	Type	Description
A	Input	Logic	First data input
B	Input	Logic	Second data input
Y	Output	Logic	EXOR output — HIGH when inputs differ
VDD	Power	Power	Supply voltage — 1.8 V
GND	Power	Power	Ground reference — 0 V

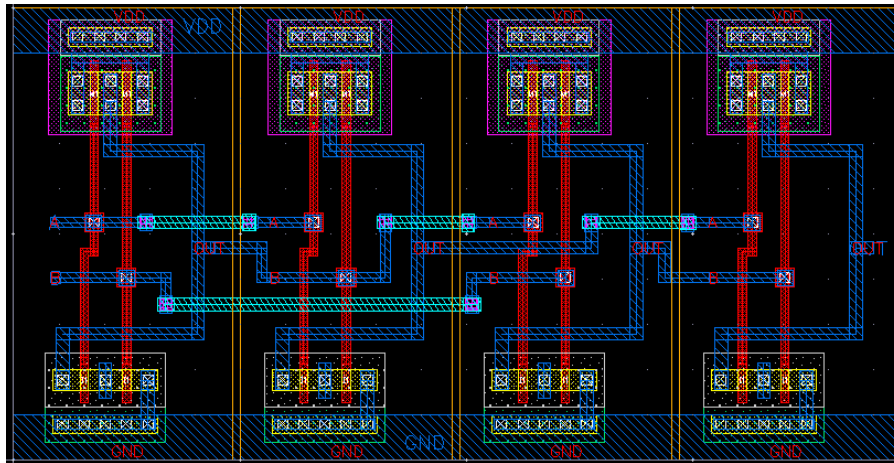
Truth Table

A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

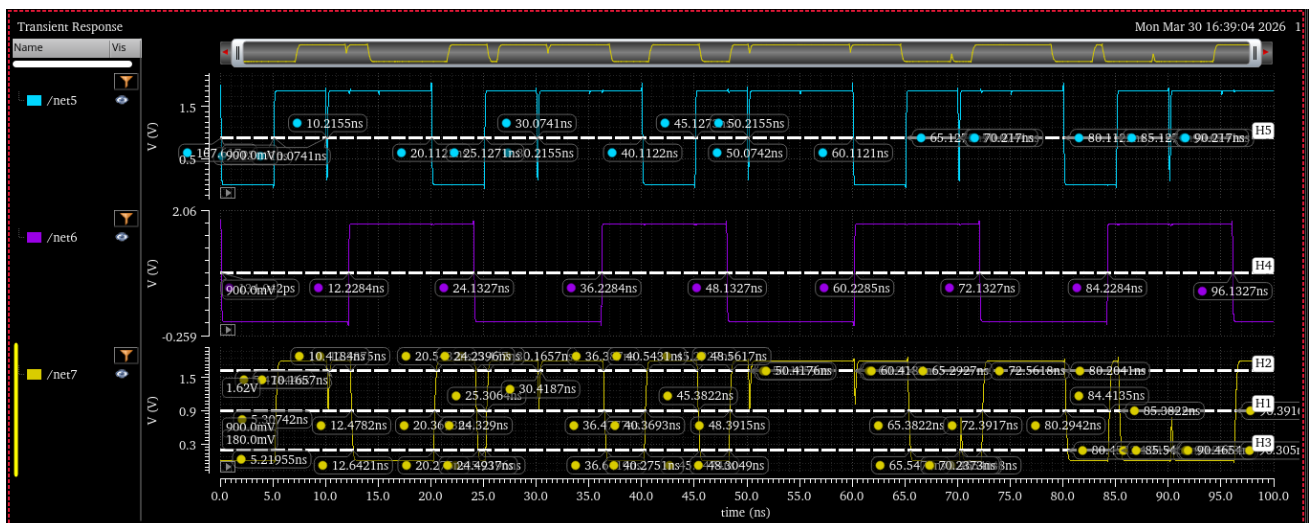
Schematic



Layout



Timing



Propagation Delay

Parameter	Symbol	Condition (one input constant)	Value	Unit
Propagation delay HL	t _{pHL}	Y: 1→0	0.2203	ns
Propagation delay LH	t _{pLH}	: 0→1	0.2289	ns
Average prop. delay	t _p	(t _{pHL} + t _{pLH}) / 2	0.22462	ns

Design Notes

- Highest propagation delay among combinational cells due to three levels of NAND logic.
- Used in 8Bit_Synchronous_Counter for the carry-chain XOR at each bit stage.
- All 4 internal NAND gates are identical to the base NAND cell — no new transistor sizing required.

7. D Flip-Flop — D_FF_with_NAND

Overview

Parameter	Value
Full cell name	D_FF_with_NAND
Boolean expression	$Q(\text{next}) = D$ [on rising CLK edge, when RST = 0]
Implementation	Master-slave configuration — two cascaded NAND-based SR latches with clock gating
Library	Gates_ED222_PY
Technology	180nm CMOS
Supply voltage (VDD)	1.8 V

Pin Configuration

Pin	Direction	Type	Description
D	Input	Logic	Data input — captured on rising CLK edge
CLK	Input	Clock	Master clock — positive edge triggered
RST	Input	Control	Synchronous active-HIGH reset — forces $Q = 0$
Q	Output	Logic	Stored output
Q'	Output	Logic	Complementary output (always $Q\text{-bar}$)
VDD	Power	Power	Supply voltage — 1.8 V
GND	Power	Power	Ground reference — 0 V

Characteristic Table

RST	D	CLK	Q (next)	Notes
1	X	↑	0	Synchronous reset — overrides D
0	0	↑	0	Captures LOW on rising edge
0	1	↑	1	Captures HIGH on rising edge
0	X	0	Q	Hold — no clock edge
0	X	1	Q	Hold — no clock edge (CLK high)

Propagation Delay

Parameter	Symbol	Condition	Value	Unit
CLK-to-Q	T_{c-Q}	CLK↑ 50% → Q: 0→1 at 50%	0.0953	ns

Design Notes

- Master-slave architecture: master latch transparent when CLK = 0, slave transparent when CLK = 1.
- RST is synchronous — takes effect on the next rising CLK edge.
- Both Q and Q' are always available.
- Used as the storage element in both 8Bit_Synchronous_Counter (8 instances) and PISO_SR (8 instances).